

Jonathan Zeng

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EDUCATION

Yale University

Expected Graduation Date: May 2028

Bachelor of Science in Computer Science and Mathematics

- **GPA:** 3.93/4.0
- **Relevant Coursework:** Intensive Algorithms, Computer Systems, Data Structures, Full Stack Web Programming, Building Game Engines, Computational Vision, Real Analysis, Discrete Mathematics, Linear Algebra

EXPERIENCE

Software Intern

May 2026 – August 2026

IBM

- Worked on the Watsonx NLP workbench for evaluating LLMs in document classification or entity extraction tasks
- Developed a backend pipeline for automated prompt refinement via user-guided input with Python and FastAPI
- Designed and built frontend webpage for data visualizations and user input using React and Carbon MCP tools
- Presented at department-wide Document Processing sprint demos three times and team-wide sprints weekly

Student Researcher

June 2025 – March 2026

Computer Graphics Group, Yale University

- Designing a faster Wavefront .OBJ parser to load and reload 3D objects using multithreading and SIMD
- Testing the effects of various C++ optimizations on performance, such as thread affinity, profile-guided optimizations, and memory-mapped files, resulting in a 41% speedup over other state-of-the-art parsers

Undergraduate Learning Assistant

January 2026 – Present

CPSC3230: Computer Systems, Yale University

- Tutoring students during weekly office hours, review sessions, and class-wide hackathon events
- Providing constructive feedback through grading weekly coding assignments, and answering discussion posts

Student Researcher

June 2023 – March 2024

I³T Research Lab, ECE Department, Duke University

- Created and analyzed a game-engine based dataset for monocular Simultaneous Localization and Mapping (SLAM) algorithms using OpenCV for Python and the Unreal game engine
- Conducted experiments on various SLAM parameters to find optimal number of features, levels, and scale factor
- Presented poster at NC Science & Engineering Fair, SNCU Research & Creativity Symposium, and lab meetings

PROJECTS

Face Extrusions on Half-edge Meshes | *D Programming Language, OpenGL*

March - April 2026

- Developed a 3D graphics app with OpenGL 4.5 and DLang to perform extrusions on half-edge data structures from imported OBJ meshes. Final project for CPSC4792: Real-Time 3D Computer Graphics.
- Implemented raycasting system, custom half-edge traversal algorithms, and camera with custom math library

Unplugged | *Swift*

February - May 2026

- Built an iOS app with a 5-person team that utilizes the Apple ScreenTime API to lock apps for productivity
- Handled backend logic for users/rooms with Vapor and PostgreSQL to create models, migrations, and controllers

Isotetris | *D Programming Language*

November - December 2025

- Designed and developed an original game/game engine featuring a 3D Tetris-like and a built-in editor
- Utilized D and SDL to create core engine for isometric rendering logic and custom resource manager for assets

Shader Lab | *JavaScript*

September 2026

- Created interactive website showcasing 3D geometry and shaders using Three.js for WebGL integration
- Wrote custom shaders using GLSL and UI layout for shader options with HTML and CSS

TECHNICAL SKILLS

Languages: C, C++, Python, D, Java, SQL, Kotlin, Lua, C#, Swift, Racket, Javascript, HTML, CSS, Markdown

Tools: Git, Github, Linux, Visual Studio, Unity, Unreal Engine, Godot, IntelliJ, Jupyter

Libraries/Frameworks: OpenGL, React, FastAPI, Flask, PyTorch, TensorFlow, SDL, OpenCV, NumPy, Matplotlib